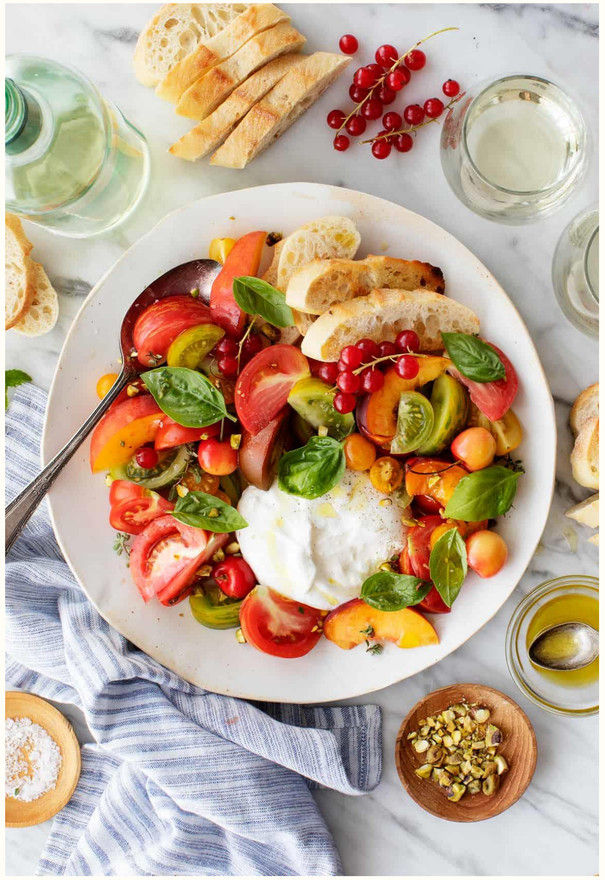
Lecture 5: R Basics Continued

Heather Mattie, PhD

September 16, 2025

Recipe of the Day!

Burrata with heirloom tomatoes and peaches



Announcements

- If you didn’t fill out the survey during week 1, please email me your GitHub username
- There will be lab this week
- Homework #1 deadline pushed to Friday, September 26
- Grab a duckie!

Office Hours and Lab

Office Hours

Day	Time	Staff	Location
Monday	10-11am	Sajia	FXB G10
Tuesday	1-2pm	Heather	Building 1, 4th floor, Room 421A
Wednesday	2:30-3:30pm	Carmen	Kresge 201 except on 10/1 where it will be held in Kresge 502
Thursday	12-1pm	Claire	Kresge 205

Lab

Day	Time	Location
Friday	11:30am-1pm	Zoom

***Lab will not be held every week**

Helpful R Resource

R functions glossary

We won't be able to cover every function included in the glossary, but it's a nice resource for the ones we will cover

Glossary

Cheat sheet of functions used in the lessons

Lesson 1 – Introduction to R

- `sqrt()` # calculate the square root
- `round()` # round a number
- `args()` # find what arguments a function takes
- `length()` # how many elements are in a particular vector
- `class()` # the class (the type of element) of an object
- `str()` # an overview of the object and the elements it contains
- `typeof` # determines the (R internal) type or storage mode of any object
- `c()` # create vector; add elements to vector
- `[ ]` # extract and subset vector
- `%in%` # to test if a value is found in a vector
- `is.na()` # test if there are missing values
- `na.omit()` # Returns the object with incomplete cases removed
- `complete.cases()`# elements which are complete cases

R Basics

R Basics Roadmap

1. Data types
2. Vectors
3. Sorting
4. Indexing
5. Basic plots
6. Programming basics

R Basics

Sorting

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sort

The `sort` function sorts a vector in **increasing** order

- It does not indicate the placement, or index, of the values once they have been sorted

```
library(dslabs)
data(murders)
sort(murders$total)
```

```
##  [1]    2    4    5    5    7    8   11   12   12   16   19   21   22   27   32
## [16]   36   38   53   63   65   67   84   93   93   97   97   99  111  116  118
## [31]  120  135  142  207  219  232  246  250  286  293  310  321  351  364  376
## [46]  413  457  517  669  805 1257
```

order

The `order` function takes a vector and returns a vector of indices

- The indices show how to sort the original vector
- It does not sort the data itself

```
x <- c(31, 4, 15, 92, 65)
sort(x)
```

```
## [1]  4 15 31 65 92
```

```
index <- order(x)
x[index]
```

```
## [1]  4 15 31 65 92
```

```
x
```

```
## [1] 31  4 15 92 65
```

```
order(x)
```

```
## [1] 2 3 1 5 4
```

order

```
library(dslabs)
data(murders)
ind <- order(murders$total)
murders$abb[ind]
```

```
##  [1] "VT" "ND" "NH" "WY" "HI" "SD" "ME" "ID" "MT" "RI" "AK" "IA" "UT" "WV" "NE"
## [16] "OR" "DE" "MN" "KS" "CO" "NM" "NV" "AR" "WA" "CT" "WI" "DC" "OK" "KY" "MA"
## [31] "MS" "AL" "IN" "SC" "TN" "AZ" "NJ" "VA" "NC" "MD" "OH" "MO" "LA" "IL" "GA"
## [46] "MI" "PA" "NY" "FL" "TX" "CA"
```

Rate example using `order`

Vector arithmetic can be especially useful for creating new variables quickly

- Example: creating and comparing murder rates for each state rather than the total number of murders in a particular state

```
head(murders)
```

```
##        state abb region population total
## 1    Alabama  AL  South    4779736   135
## 2     Alaska  AK   West     710231    19
## 3    Arizona  AZ   West    6392017   232
## 4   Arkansas  AR  South    2915918    93
## 5 California  CA   West   37253956  1257
## 6   Colorado  CO   West    5029196    65
```

```
murder_rate <- murders$total / murders$population * 100000

murders$state[order(murder_rate)]
```

```
##  [1] "Vermont"              "New Hampshire"        "Hawaii"
##  [4] "North Dakota"         "Iowa"                 "Idaho"
##  [7] "Utah"                 "Maine"                "Wyoming"
## [10] "Oregon"               "South Dakota"         "Minnesota"
## [13] "Montana"              "Colorado"             "Washington"
## [16] "West Virginia"        "Rhode Island"         "Wisconsin"
## [19] "Nebraska"             "Massachusetts"        "Indiana"
## [22] "Kansas"               "New York"             "Kentucky"
## [25] "Alaska"               "Ohio"                 "Connecticut"
## [28] "New Jersey"           "Alabama"              "Illinois"
## [31] "Oklahoma"             "North Carolina"       "Nevada"
## [34] "Virginia"             "Arkansas"             "Texas"
## [37] "New Mexico"           "California"           "Florida"
## [40] "Tennessee"            "Pennsylvania"         "Arizona"
## [43] "Georgia"              "Mississippi"          "Michigan"
## [46] "Delaware"             "South Carolina"       "Maryland"
## [49] "Missouri"             "Louisiana"            "District of Columbia"
```

rank

For any given vector, the **rank** function gives you a vector with the rank of the first entry, second entry, etc. of the vector

```
x <- c(31, 4, 15, 92, 65)
rank(x)
```

```
## [1] 3 1 2 5 4
```

Comparing sorting functions

original	sort	order	rank
31	4	2	3
4	15	3	1
15	31	1	2
92	65	5	5
65	92	4	4

Summary

- The `sort` function sorts a vector in increasing order
- The `order` function returns a vector of indicies that sort the original vector
- The `rank` function returns a vector with the rank of each entry when the original vector is sorted

R Basics

Indexing

Heather Mattie, PhD

`murders` dataset

```
head(murders)
```

```
##        state abb region population total
## 1    Alabama  AL  South    4779736   135
## 2     Alaska  AK   West     710231    19
## 3    Arizona  AZ   West    6392017   232
## 4   Arkansas  AR  South    2915918    93
## 5 California  CA   West   37253956  1257
## 6   Colorado  CO   West    5029196    65
```

max and which.max

The **max** function returns the maximum value in a vector

The **which.max** function returns the index of the maximum value

```
max(murders$total)
```

```
## [1] 1257
```

```
i_max <- which.max(murders$total)
i_max
```

```
## [1] 5
```

```
murders$state[i_max]
```

```
## [1] "California"
```

Subsetting with logicals

Logical statements (logicals) can be used to index vectors

- Return TRUE or FALSE for each entry in a vector
- TRUE values are coded as 1
- FALSE values are coded as 0

```
murder_rate <- murders$total / murders$population * 100000
ind <- murder_rate < 0.71
ind
```

```
##  [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
## [13] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [25] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE  TRUE FALSE
## [37] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
## [49] FALSE FALSE FALSE
```

```
sum(ind)
```

```
## [1] 5
```

```
murders$state[ind]
```

```
## [1] "Hawaii"        "Iowa"          "New Hampshire" "North Dakota"
## [5] "Vermont"
```

Logical operators

Logical operators are symbols or words used to connect two or more expressions

- Result in a TRUE or FALSE
- AND = **&**
- OR = **|**
- NOT = **!**

```
TRUE & TRUE
```

```
## [1] TRUE
```

```
TRUE & FALSE
```

```
## [1] FALSE
```

```
FALSE & FALSE
```

```
## [1] FALSE
```

```
west <- murders$region == "West"
safe <- murder_rate <= 1
ind <- safe & west
murders$state[ind]
```

```
## [1] "Hawaii"  "Idaho"   "Oregon"  "Utah"    "Wyoming"
```

Logical operators

Logical operators are symbols or words used to connect two or more expressions

- Result in a TRUE or FALSE
- AND = **&**
- OR = **|**
- NOT = **!**

```
ind <- safe | west
murders$state[ind]
```

```
##  [1] "Alaska"        "Arizona"       "California"    "Colorado"
##  [5] "Hawaii"        "Idaho"         "Iowa"          "Maine"
##  [9] "Minnesota"     "Montana"       "Nevada"        "New Hampshire"
## [13] "New Mexico"    "North Dakota"  "Oregon"        "South Dakota"
## [17] "Utah"          "Vermont"       "Washington"    "Wyoming"
```

```
ind <- safe & !west
murders$state[ind]
```

```
## [1] "Iowa"          "Maine"         "Minnesota"     "New Hampshire"
## [5] "North Dakota"  "South Dakota"  "Vermont"
```

which

The **which** function indicates which entries of a logical vector are TRUE

```
ind <- which(murders$state == "California")
ind # this is the index that matches the California entry
```

```
## [1] 5
```

```
murder_rate[ind]
```

```
## [1] 3.374138
```

match

The `match` function indicates which indices of a second vector match each of the entries of a first vector

```
ind <- match(c("New York", "Florida", "Texas"), murders$state)
ind
```

```
## [1] 33 10 44
```

```
match(c("Boston", "Dakota", "Washington"), murders$state)
```

```
## [1] NA NA 48
```

```
murder_rate[ind]
```

%in%

The `%in%` function indicates whether or not each element in a vector is contained in another vector

```
c("Boston", "Dakota", "Washington") %in% murders$state
```

```
## [1] FALSE FALSE  TRUE
```

Summary

- Vectors can be subset with logical statements
- Logical operators (`&`, `|`, `!`) can connect two or more logical statements
- The `which`, `match` and `%in%` functions can be used to index where TRUE values are in a vector

R Basics

Basic Plots

Heather Mattie, PhD

Base R plots

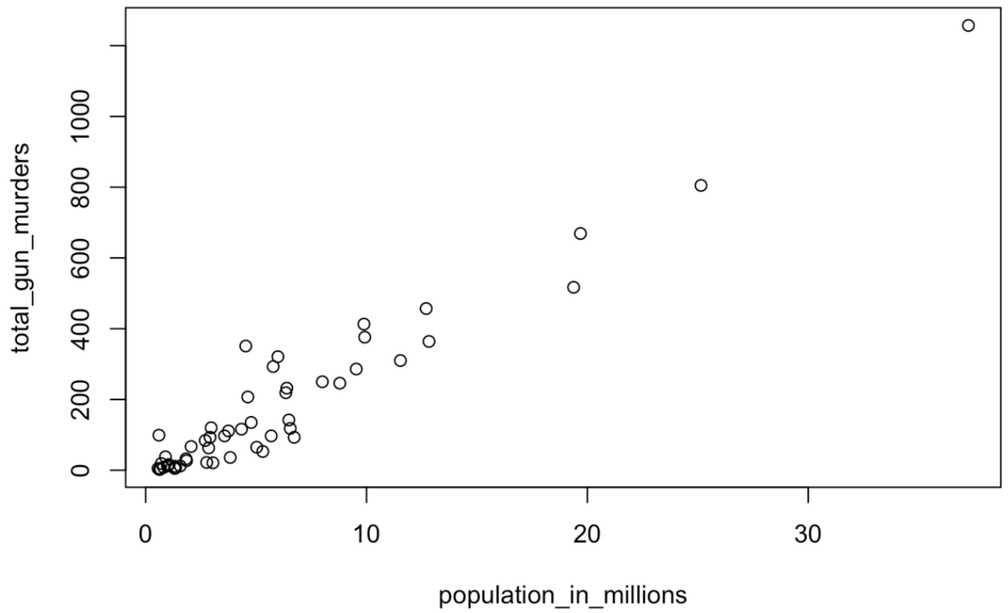
Plots can be effective and efficient ways to visualize data

- The type of plot depends on the data type(s)
- Exploratory data visualization is an important part of the data science project pipeline
- These plots don’t require installing and loading any other R packages, but are rudimentary

Scatter plots

Scatter plots show the relationship between two continuous (numeric) variables

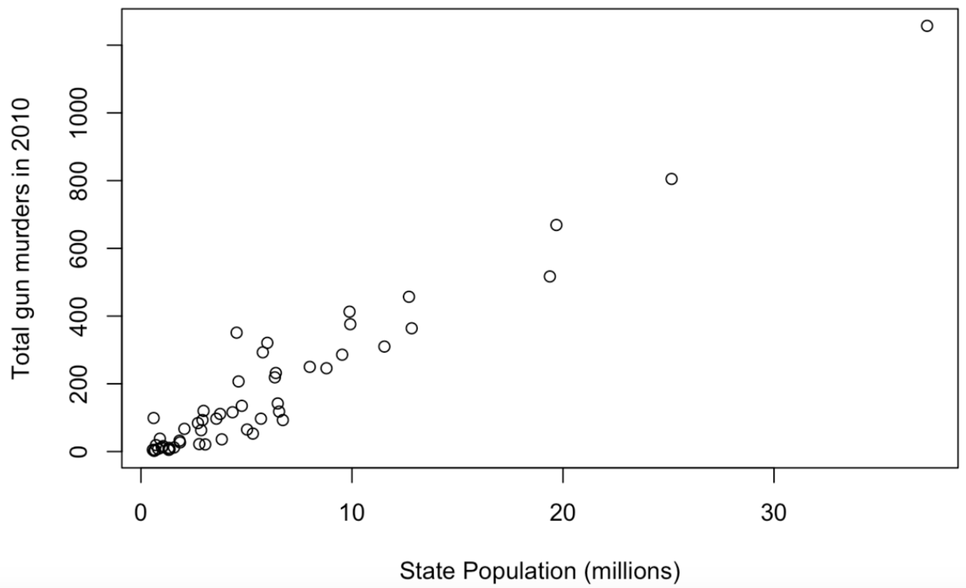
```
population_in_millions <- murders$population/10^6
total_gun_murders <- murders$total
plot(x = population_in_millions, y = total_gun_murders)
```



Scatter plots

Scatter plots show the relationship between two continuous (numeric) variables

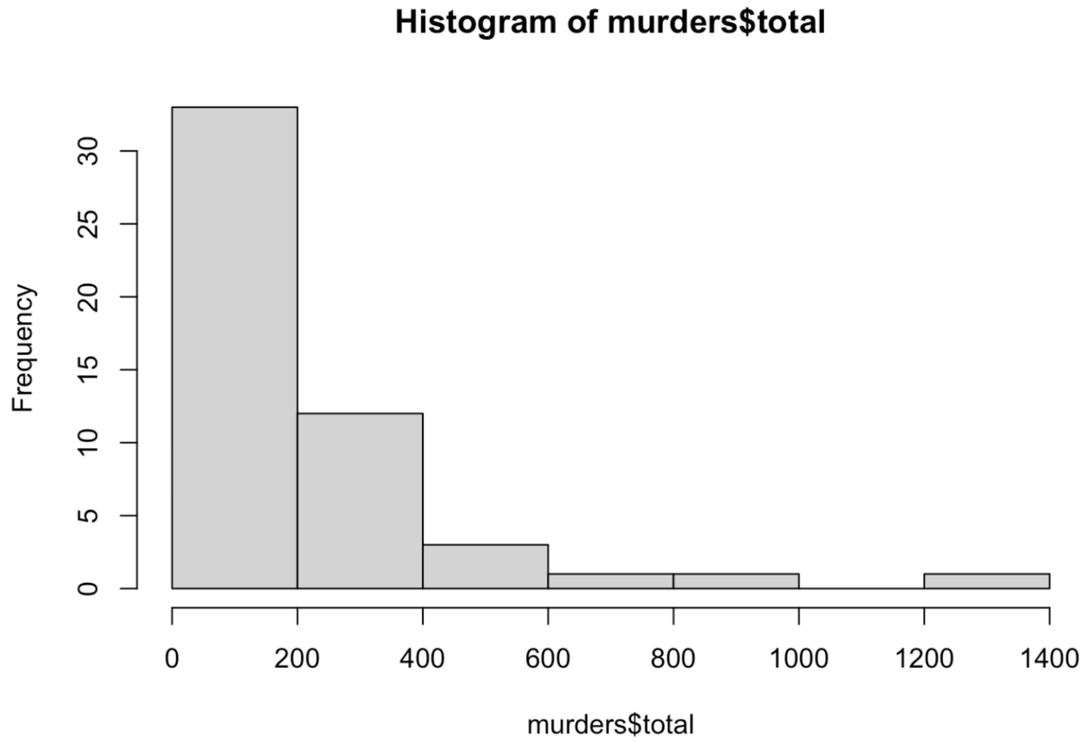
```
population_in_millions <- murders$population/10^6
total_gun_murders <- murders$total
plot(x = population_in_millions, y = total_gun_murders,
     xlab = "State Population (millions)",
     ylab = "Total gun murders in 2010")
```



Histograms

Histograms plot the distribution of values for one continuous (numeric) variable

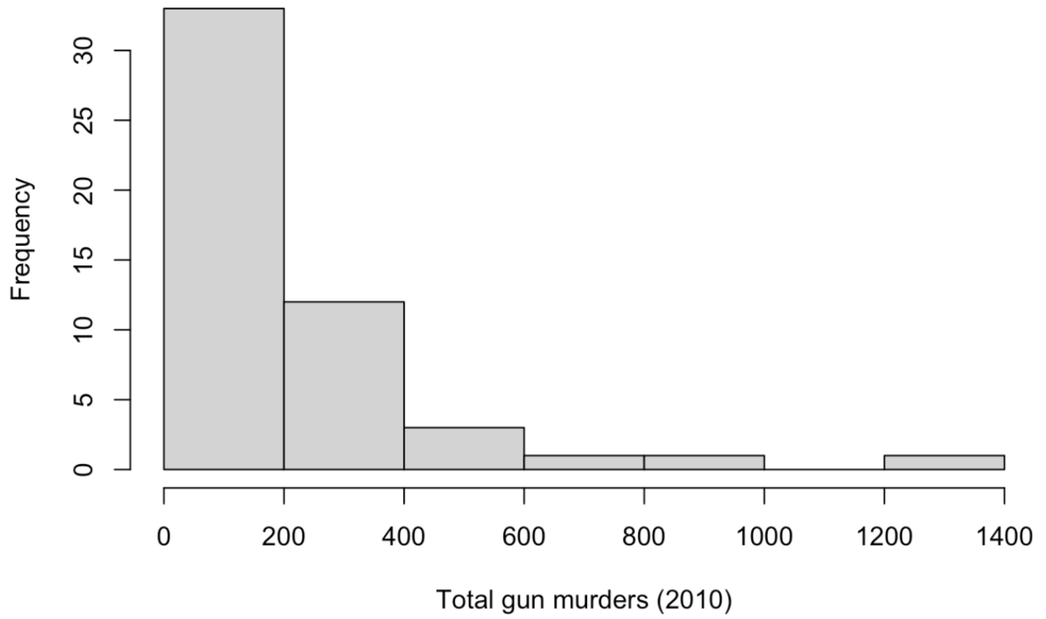
```
hist(murders$total)
```



Histograms

Histograms plot the distribution of values for one continuous (numeric) variable

```
hist(murders$total, xlab = "Total gun murders (2010)",
     main = "")
```

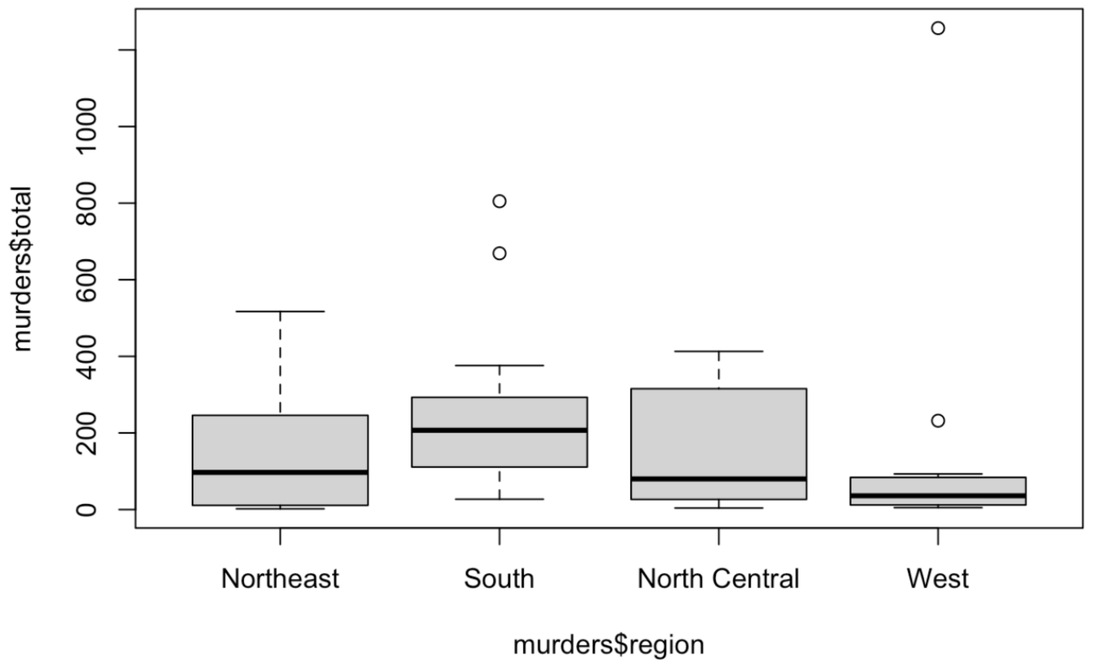


Boxplots

Boxplots display the distribution of a continuous variable based on a five-number summary

- Minimum, first quartile (Q1), median, third quartile (Q3), and maximum
- Display outliers if they exist
- Can plot multiple boxplots in one plot

```
boxplot(murders$total ~ murders$region)
```

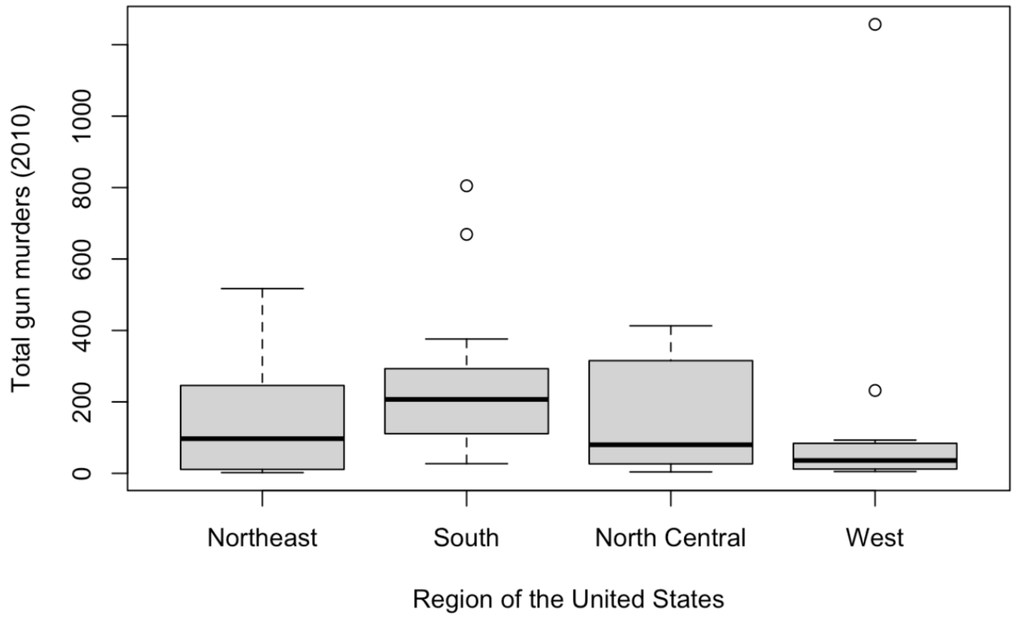


Boxplots

Boxplots display the distribution of a continuous variable based on a five-number summary

- Minimum, first quartile (Q1), median, third quartile (Q3), and maximum
- Display outliers if they exist
- Can plot multiple boxplots in one plot

```
boxplot(murders$total ~ murders$region,
        xlab = "Region of the United States",
        ylab = "Total gun murders (2010)")
```

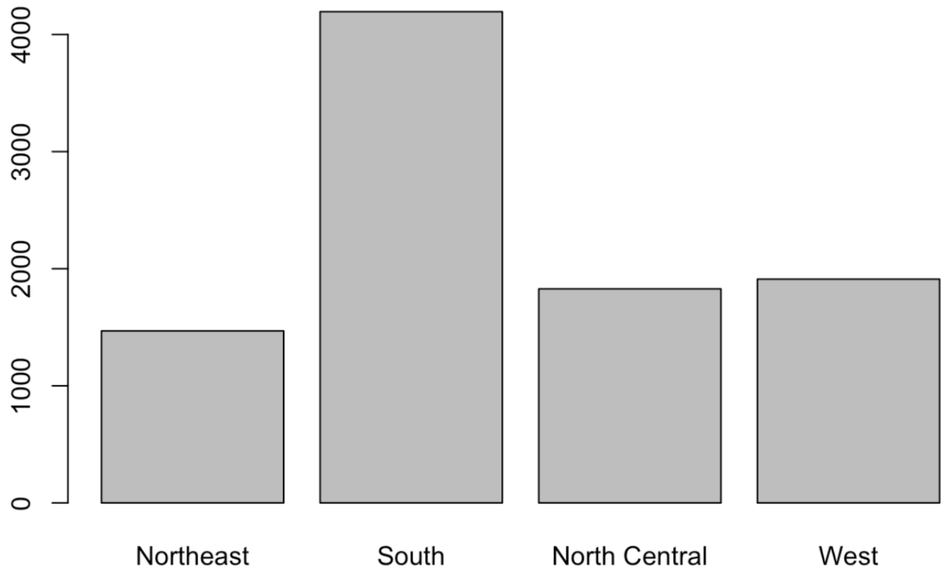


Barplots

Barplots show the relationship between a numeric variable and categorical variable

- Each bar represents a category
- The length of a bar represents its numeric value

```
data <- data.frame(
  name = c("Northeast", "South", "North Central", "West"),
  value = c(1469, 4195, 1828, 1911))
barplot(height = data$value, names = data$name)
```

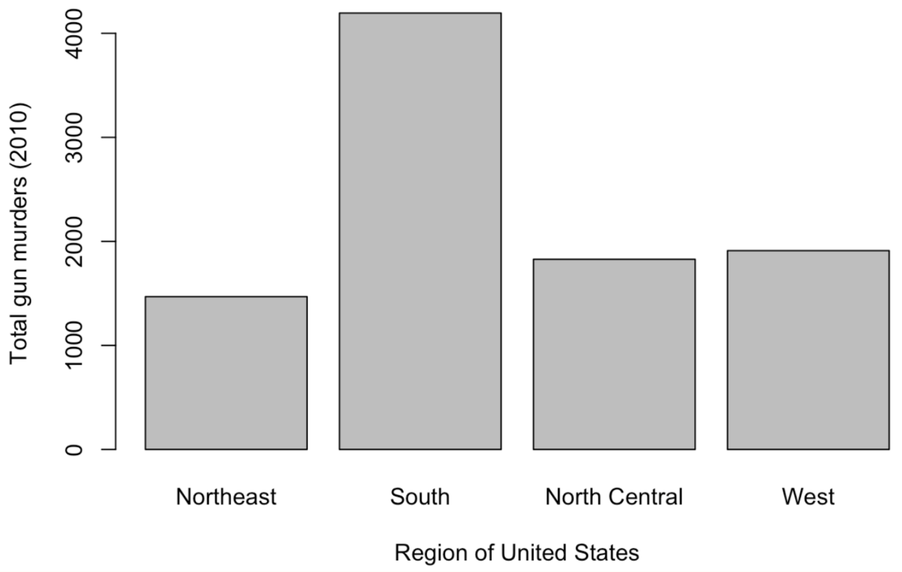


Barplots

Barplots show the relationship between a numeric variable and categorical variable

- Each bar represents a category
- The length of a bar represents its numeric value

```
data <- data.frame(
  name = c("Northeast", "South", "North Central", "West"),
  value = c(1469, 4195, 1828, 1911))
barplot(height = data$value, names = data$name,
        xlab = "Region of United States",
        ylab = "Total gun murders (2010)")
```



Summary

- Plots are a necessary and effective tool to gain insights into data
- Base R plot functions can be used to create rudimentary plots
- Scatter plots, histograms, boxplots, and barplots are commonly used to display data

R Basics

Programming Basics

Heather Mattie, PhD

Programming basics

Programming is the process of creating a set of instructions (code) that a computer can understand and execute to perform tasks or solve problems

- Conditional expressions
- For loops
- Defining functions

Conditional expressions

Conditional expressions (if-else statements) execute different blocks of code based on whether a condition is true or false

```
a <- 0

if(a != 0){
  print(1/a)
} else{
  print("No reciprocal for 0.")
}
```

```
## [1] "No reciprocal for 0."
```

```
a <- 2

if(a != 0){
  print(1/a)
} else{
  print("No reciprocal for 0.")
}
```

```
## [1] 0.5
```

Conditional expressions

Conditional expressions (if-else statements) execute different blocks of code based on whether a condition is true or false

```
a <- 0
ifelse(a > 0, 1/a, NA)
```

```
## [1] NA
```

```
a <- c(0, 1, 2, -4, 5)
result <- ifelse(a > 0, 1/a, NA)
result
```

```
## [1]  NA 1.0 0.5  NA 0.2
```

Conditional expressions

Conditional expressions (if-else statements) execute different blocks of code based on whether a condition is true or false

```
library(dslabs)
data(murders)
murder_rate <- murders$total/murders$population*100000
ind <- which.min(murder_rate)

ifelse(murder_rate[ind] < 0.5, murders$state[ind], print
("Minumum murder rate is not that low."))
```

```
## [1] "Vermont"
```

```
ifelse(murder_rate[ind] < 0.25, murders$state[ind], print
("Minumum murder rate is not that low."))
```

```
## [1] "Minumum murder rate is not that low."
```

For loops

For loops repeatedly execute a block of code a specific number of times or until a certain condition is met

- Used for iteration
- Used when number of repetitions is known in advance

```
for(i in 1:5){
  print(i)
}
```

```
## [1] 1
## [1] 2
## [1] 3
## [1] 4
## [1] 5
```

For loops

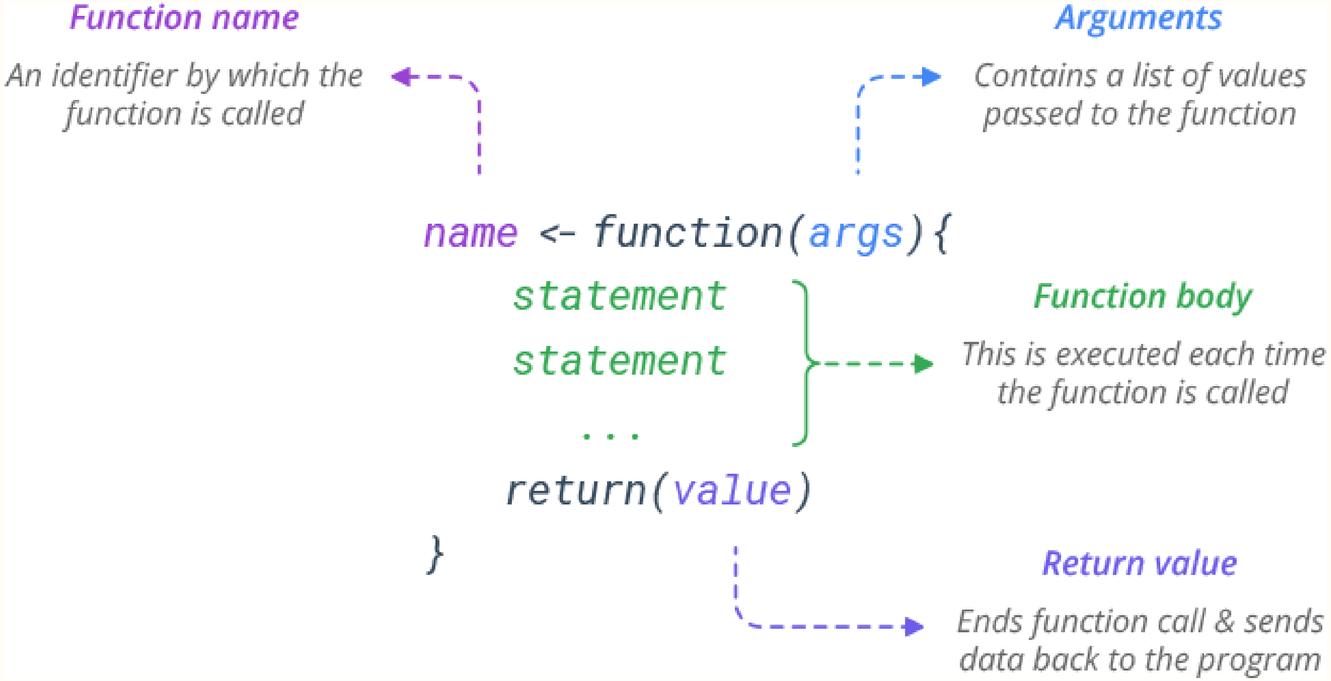
```r
# Filter to Northeast region
northeast_data <- murders[murders$region == "Northeast", ]

# Use a for loop to print murder rates
for (i in 1:nrow(northeast_data)) {
  state_name <- northeast_data$state[i]
  rate <- (northeast_data$total[i] / northeast_data$population[i]) * 100000
  cat(state_name, "has a murder rate of", round(rate, 2), "per 100,000 people.\n")
}
```

```
## Connecticut has a murder rate of 2.71 per 100,000 people.
## Maine has a murder rate of 0.83 per 100,000 people.
## Massachusetts has a murder rate of 1.8 per 100,000 people.
## New Hampshire has a murder rate of 0.38 per 100,000 people.
## New Jersey has a murder rate of 2.8 per 100,000 people.
## New York has a murder rate of 2.67 per 100,000 people.
## Pennsylvania has a murder rate of 3.6 per 100,000 people.
## Rhode Island has a murder rate of 1.52 per 100,000 people.
## Vermont has a murder rate of 0.32 per 100,000 people.
```

Defining functions

Custom functions can be created and used in R

- If no such function exists
- If your code performs the same task more than once



Defining functions

Custom functions can be created and used in R

- If no such function exists
- If your code performs the same task more than once

```
# Anatomy of a function

name <- function(x){
  code to run
  output
}
```

```
avg <- function(x){
  s <- sum(x)
  n <- length(x)
  s/n
}

x <- 1:100
avg(x)
```

```
## [1] 50.5
```

Defining functions

Custom functions can be created and used in R

- If no such function exists
- If your code performs the same task more than once

```
compute_murder_rate <- function(total, population) {
  rate <- (total / population) * 100000
  return(rate)
}

# Get total murders and population for New York
ny_total <- murders$total[murders$state == "New York"]
ny_pop <- murders$population[murders$state == "New York"]

# Compute rate
compute_murder_rate(ny_total, ny_pop)
```

```
## [1] 2.66796
```

Summary

- If-else statements can be used to execute different parts of code based on a condition
- For loops repeatedly execute code for a set number of times
- Custom functions can be created and used in R